

Regression Analysis

Group1

統計所 R26121023 張伊萱
統計所 R26124055 施文千
統計所 R26124013 李品嫻

CONTENTS

張伊萱

- 01. Introduction
- 02. Simple linear regression
- 03. Estimation and hypothesis testing

施文千

- 04. Multiple linear regression
- 05. Diagnostics
- 06. Hypothesis testing
- 07. Loess

李品嫻

- 08. Qualitative Predictors
- 09. Interaction Model
- 10. Model Selection
- 11. Selection Procedure
- 12. Multicollinearity

01.

Introduction



The purpose of regression analysis

The main focus is on exploring the relationship between the explanatory variable (X) and the response variable (Y). By establishing a regression model, researchers can infer and predict the variable of interest (Y).



02.

Simple linear regression



Simple linear regression

- Using a single explanatory variable x to predict a response variable y

$$Y_i = \beta_0 + \beta_1 x_i + \varepsilon_i, \quad i = 1, 2, \dots, n$$

其中 Y_i represents the observed value of the response variable for the i th sample.

x_i represents the observed value of the explanatory variable for the i th sample.

β_0 denotes the intercept term in the regression equation.

β_1 denotes the slope term in the regression equation.

ε_i represents the error term for the i th sample

n denotes the sample size.

Assumptions of Simple Linear Regression

$$Y_i = \beta_0 + \beta_1 x_i + \varepsilon_i, \quad i = 1, 2, \dots, n$$
$$\varepsilon_i \stackrel{\text{iid}}{\sim} N(0, \sigma^2)$$

- Linear Assumption: There exists a linear relationship between Y_i and x_i .
- ε_i is a random variable with $E(\varepsilon_i) = 0$ and $\text{Var}(\varepsilon_i) = \sigma^2$, for $i = 1, \dots, n$
- ε_i and ε_j uncorrelated. $\Rightarrow \text{cov}(\varepsilon_i, \varepsilon_j) = 0, \forall i \neq j$
- **Normality Assumption: The error terms follow a normal distribution, which implies Y_i follow a normal distribution.

$$\rightarrow Y_i \stackrel{\text{indep}}{\sim} N(\beta_0 + \beta_1 x_i, \sigma^2)$$

population regression equation

population regression equation:

$$E(Y_i) = \beta_0 + \beta_1 x_i, \quad i = 1, 2, \dots, n$$

- The true value Y_i must be added with an individual error term ε_i .
 $\Rightarrow Y_i = E(Y_i) + \varepsilon_i$
- The population regression equation is unknown, the sample regression equation $\hat{Y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i$ is used to estimate the population regression equation $E(Y_i) = \beta_0 + \beta_1 x_i$.
- Residual $e_i = Y_i - \hat{Y}_i \xrightarrow{\text{估計}}$ error term $\varepsilon_i = Y_i - E(Y_i)$

03.

Estimation and hypothesis testing



估計係數 方法

最小平方法
(Least squares method)

目的: 求取最小化殘差平方和的點估計量

最大概似估計
(Maximum Likelihood Estimation)

目的: 求取最大化概似函數的點估計量



Least Squares Method

(Note) **Normality assumption not required**

$$\min L = \sum_{i=1}^n e_i^2 = \sum_{i=1}^n (y_i - \hat{y}_i)^2 = \sum_{i=1}^n (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i)^2$$

$$\begin{cases} \frac{\partial}{\partial \hat{\beta}_0} L = -2 \sum_{i=1}^n (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i) = 0 \\ \frac{\partial}{\partial \hat{\beta}_1} L = -2 \sum_{i=1}^n (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i) x_i = 0 \end{cases} \Rightarrow \begin{cases} \hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x} \\ \hat{\beta}_1 = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2} = \frac{S_{xy}}{S_{xx}} \end{cases}$$

$$\text{檢查: } H = \det \begin{vmatrix} \frac{\partial^2}{\partial (\hat{\beta}_0)^2} L & \frac{\partial^2}{\partial \hat{\beta}_0 \partial \hat{\beta}_1} L \\ \frac{\partial^2}{\partial \hat{\beta}_0 \partial \hat{\beta}_1} L & \frac{\partial^2}{\partial (\hat{\beta}_1)^2} L \end{vmatrix} > 0 \text{ 且 } \frac{\partial^2}{\partial (\hat{\beta}_0)^2} L > 0$$

Maximum Likelihood Estimation

(Note) **require** the normality assumption.

$$\max L(\beta_0, \beta_1, \sigma^2) = \prod_{i=1}^n \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{1}{2}\left(\frac{y_i - \beta_0 - \beta_1 x_i}{\sigma}\right)^2}$$

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$$

$$\hat{\beta}_1 = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2} = \frac{S_{xy}}{S_{xx}}$$

$$\hat{\sigma}_{MLE}^2 = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2 = \frac{1}{n} \sum_{i=1}^n e_i^2 = \frac{SSE}{n} \text{ (biased estimator)}$$

$$\Rightarrow MSE = \frac{SSE}{n-2} \xrightarrow{\text{估計}} \sigma^2$$

Interval Estimation-1

- $(1 - \alpha)100\%$ confidence interval for β_0 :

- point estimator: $\hat{\beta}_0 \rightarrow \beta_0$

- confidence interval: $\hat{\beta}_0 \pm t_{\frac{\alpha}{2}}(n - 2) \sqrt{MSE \left[\frac{1}{n} + \frac{(\bar{X})^2}{S_{xx}} \right]}$

- $(1 - \alpha)100\%$ confidence interval for β_1 :

- point estimator: $\hat{\beta}_1 \rightarrow \beta_1$

- confidence interval: $\hat{\beta}_1 \pm t_{\frac{\alpha}{2}}(n - 2) \sqrt{\frac{MSE}{S_{xx}}}$

Interval Estimation-2

- When $X = a$, the $(1 - \alpha)100\%$ confidence interval for $E(Y) = \beta_0 + \beta_1 \times a$:

- point estimator: $\hat{Y} = \beta_0 + \beta_1 \times a \rightarrow E(Y)$

- confidence interval: $\hat{Y} \pm t_{\frac{\alpha}{2}}(n - 2) \sqrt{MSE \left[\frac{1}{n} + \frac{(a - \bar{X})^2}{S_{xx}} \right]}$

- When $X = a$, the $(1 - \alpha)100\%$ prediction interval for $Y = E(Y) + \varepsilon$:

- point estimator: $\hat{Y} = \beta_0 + \beta_1 \times a \rightarrow Y$

- prediction interval: $\hat{Y} \pm t_{\frac{\alpha}{2}}(n - 2) \sqrt{MSE \left[1 + \frac{1}{n} + \frac{(a - \bar{X})^2}{S_{xx}} \right]}$

Hypothesis testing

- Hypothesis testing for the population regression slope β_1 :

① $\begin{cases} H_0: \beta_1 = b \\ H_1: \beta_1 \neq b \end{cases}$ (通常檢定 $b=0$)

② Setting the significance level α

③ Test statistic: $T = \frac{\hat{\beta}_1 - b}{\sqrt{\frac{MSE}{Sxx}}} = \frac{\hat{\beta}_1 - b}{se(\hat{\beta}_1)} \sim t(n-2)$

④ Rejection region:

$$RR = \left\{ T \leq -t_{\frac{\alpha}{2}}(n-2) \text{ 或 } T \geq t_{\frac{\alpha}{2}}(n-2) \right\}$$

⑤ calculate $T_0 = \frac{\hat{\beta}_1 - b}{\sqrt{\frac{MSE}{Sxx}}}$

⑥ Decision rule:

If $T_0 \in RR$, reject H_0 ;

If $T_0 \notin RR$, do not reject H_0

Example

假設我們認為膽固醇含量與每周運動時間存在一定程度的線性關係，想要知道這兩者之間的關係，以下記錄了某12位同學之膽固醇含量 (mg/dl) 與每周運動時間的資料並建立迴歸模型。

膽固醇含量 (Y)	224	216	212	204	212	188	192	204	192	188	184	176
運動時間(X)	0	0	0	2	2	3	3	3	5	5	8	8

假設迴歸模型： $Y_i = \beta_0 + \beta_1 x_i + \varepsilon_i$, $i = 1, 2, \dots, 12$
 $\varepsilon_i \stackrel{\text{iid}}{\sim} N(0, \sigma^2)$

Example

	Estimate	Std. Error	t value	Pr(> t)
Intercept	214.7826	3.0285	70.92	7.58E-15
運動時間(X)	-4.7536	0.7188	-6.613	5.98E-05
Residual standard error: 6.676 on 10 degrees of freedom Multiple R-squared: 0.8139, Adjusted R-squared: 0.7953 F-statistic: 43.73 on 1 and 10 DF, p-value: 5.981e-05				

- 膽固醇含量與每周運動時間存在線性關係
- 估計迴歸方程式： $\hat{Y}_i = 214.7826 - 4.7536x_i$
- 假設今天有個同學聲稱每周運動18小時，則可計算出
 $\hat{Y} = 214.7826 - 4.7536 \times 18 = 129.2177$ (mg/dl)
- $R^2 = 0.8139 \Rightarrow$ 模型可解釋81.39%膽固醇含量的變異

04.

Multiple linear regression



Multiple linear Regression

Let the model be

$$y_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \cdots + \beta_p x_{ip} + \varepsilon_i, \quad i = 1, \dots, n.$$

Three assumptions are

1. $E(\varepsilon_i) = 0, \quad \forall i$
2. $\text{var}(\varepsilon_i) = \sigma^2, \quad \forall i$
3. $\text{cov}(\varepsilon_i, \varepsilon_j) = 0, \quad \forall i \neq j$

Multiple linear Regression

For n observations we can use matrix form to express the formula:

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon},$$

where

$$\mathbf{y} = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix}, \quad \boldsymbol{\beta} = \begin{bmatrix} \beta_1 \\ \beta_2 \\ \vdots \\ \beta_n \end{bmatrix}, \quad \text{and } \boldsymbol{\varepsilon} = \begin{bmatrix} \varepsilon_1 \\ \varepsilon_2 \\ \vdots \\ \varepsilon_n \end{bmatrix}$$

Multiple linear Regression

The matrix for p predictors:

$$\mathbf{X} = \begin{bmatrix} 1 & x_{11} & \cdots & x_{1p} \\ \vdots & & \ddots & \vdots \\ 1 & x_{n1} & \cdots & x_{np} \end{bmatrix}$$

Three assumptions are

1. $E(\varepsilon) = \mathbf{0}$
2. By Assumptions 2 and 3, $\text{cov}(\varepsilon) = \sigma^2 \mathbf{I}_{n \times n}$

we assume that $n > p + 1$ and $\text{rank}(\mathbf{X}) = p + 1$.

Multiple linear Regression

The least squares estimators for β by minimizing

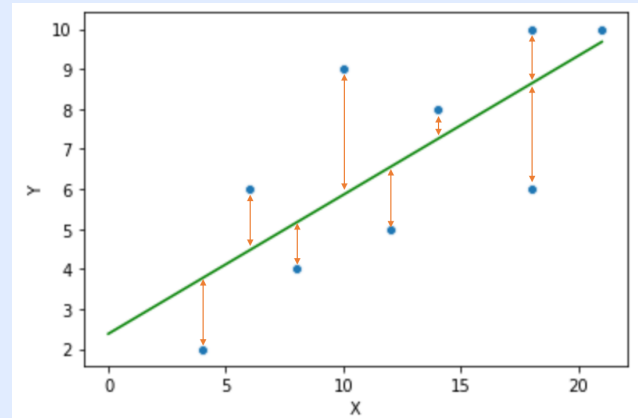
$$\sum_{i=1}^n \hat{\varepsilon}_i^2 = \sum_{i=1}^n (y_i - \hat{y}_i)^2 = \sum_{i=1}^n (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_{i1} - \dots - \hat{\beta}_p x_{ip})^2$$

Then, the estimator in matrix form is

$$\hat{\beta} = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y}$$

The properties of $\hat{\beta}$:

1. $E(\mathbf{y}) = \mathbf{X}\beta$, then $\hat{\beta}$ is unbiased.
2. $cov(\hat{\beta}) = (\mathbf{X}^T \mathbf{X})^{-1} \sigma^2$



Multiple linear Regression

- $E(\mathbf{y}) = \mathbf{X}\beta$, then $\hat{\beta}$ is unbiased.

$$E(\hat{\beta}) = E(\beta + (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \varepsilon) = \beta + (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top E(\varepsilon) = \beta$$

- $cov(\hat{\beta}) = (\mathbf{X}^\top \mathbf{X})^{-1} \sigma^2$

$$\begin{aligned} cov(\hat{\beta}) &= E \left[\left(\hat{\beta} - E(\hat{\beta}) \right) \left(\hat{\beta} - E(\hat{\beta}) \right)^\top \right] = E \left[\left((\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \varepsilon \right) \left((\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \varepsilon \right)^\top \right] \\ &= E \left[(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \varepsilon \varepsilon^\top \mathbf{X} (\mathbf{X}^\top \mathbf{X})^{-1} \right] = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top E(\varepsilon \varepsilon^\top) \mathbf{X} (\mathbf{X}^\top \mathbf{X})^{-1} \\ &= (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \sigma^2 \mathbf{I}_{p+1} \mathbf{X} (\mathbf{X}^\top \mathbf{X})^{-1} = \sigma^2 (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{X} (\mathbf{X}^\top \mathbf{X})^{-1} \\ &= \sigma^2 (\mathbf{X}^\top \mathbf{X})^{-1} \end{aligned}$$

Multiple linear Regression

When adding or removing a variable X_j , the R_{adj}^2 will increase, decrease or don't change, so we can use t ratio value.

$$t_j = \frac{\hat{\beta}_j}{SE(\hat{\beta}_j)}$$

$$\begin{cases} R_{adj}^2 \text{ increase, if } |t_j| > 1 \\ R_{adj}^2 \text{ don't change, if } |t_j| = 1 \\ R_{adj}^2 \text{ decrease, if } |t_j| < 1. \end{cases}$$

05.

Hypothesis testing



Hypothesis testing

In simple linear regression ,

$$y_i = \beta_0 + \beta_1 x_i + \varepsilon_i , i = 1, \dots, n$$

where $\varepsilon_i \sim N(0, \sigma^2)$

$$H_0 : \beta_1 = 0$$

$$H_a : \beta_1 \neq 0$$

Hypothesis testing

Some properties

1. $\widehat{\beta}_1 \sim N(\beta_1, \frac{\sigma^2}{\sum_{i=1}^n (x_i - \bar{x})^2})$
2. $s^2 = \frac{SSE}{n-2}$, then we get $\frac{(n-2)}{\sigma^2} s^2 \sim \chi_{(n-2)}^2$
3. $\widehat{\beta}_1$ and s^2 are independent

Hypothesis testing

$$H_0 : \beta_1 = 0$$

$$H_a : \beta_1 \neq 0$$

Full model: $y_i = \beta_0 + \beta_1 x_i + \varepsilon_i$

Reduced model: $y_i = \beta_0 + \varepsilon_i$

Test statistics

$$t_0 = \frac{\widehat{\beta}_1 - 0}{s \sqrt{\sum_{i=1}^n (x_i - \bar{x})^2}} \sim t_{(n-2)}$$

Hypothesis testing

In multiple linear regression ,

$$y_i = \beta_0 + \beta_1 x_1 + \cdots + \beta_p x_p + \varepsilon_i, i = 1, \dots, n$$

where $\varepsilon_i \sim N(0, \sigma^2)$

$$H_0 : \beta_1 = \beta_2 = \cdots = 0$$

$$H_a : \text{at least one } \beta_j \neq 0, j = 1, \dots, p$$

Hypothesis testing

$$H_0 : \beta_1 = \beta_2 = \dots = 0$$

$$H_a : \text{at least one } \beta_j \neq 0, j = 1, \dots, p$$

$$\text{Full model: } y_i = \beta_0 + \beta_1 x_1 + \dots + \beta_p x_p + \varepsilon_i$$

$$\text{Reduced model: } y_i = \beta_0 + \varepsilon_i$$

Hypothesis testing

	SS	DF	MS	F
SSR	$\sum (\hat{y} - \bar{y})^2$	p	$\frac{SSR}{p}$	$\frac{MSR}{MSE}$
SSE	$\sum (y - \hat{y})^2$	$n - p - 1$	$\frac{SSE}{n - p - 1}$	
SST	$\sum (y - \bar{y})^2$	$n - 1$		

Hypothesis testing

In multiple linear regression ,

$$y_i = \beta_0 + \beta_1 x_1 + \cdots + \beta_4 x_4 + \varepsilon_i , i = 1, \dots, 4$$

where $\varepsilon_i \sim N(0, \sigma^2)$

$$H_0 : \beta_2 = \beta_3 = 0$$

$$H_a : \text{at least one } \beta_j \neq 0, j = 2, 3$$

Hypothesis testing

$$H_0 : \beta_2 = \beta_3 = 0$$

$$H_a : \text{at least one } \beta_j \neq 0, j = 2, 3$$

Full model: $y_i = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + \beta_4 x_4 + \varepsilon_i$

Reduced model: $y_i = \beta_0 + \beta_1 x_1 + \beta_4 x_4 + \varepsilon_i$

Test statistics

$$F = \frac{(SSE_R - SSE_F)/\Delta p}{MSE_F}$$

Example

```
data(mtcars)  
head(mtcars, 10)
```

	mpg	cyl	disp	hp	drat	wt	qsec	vs	am	gear	carb
Mazda RX4	21.0	6	160.0	110	3.90	2.620	16.46	0	1	4	4
Mazda RX4 Wag	21.0	6	160.0	110	3.90	2.875	17.02	0	1	4	4
Datsun 710	22.8	4	108.0	93	3.85	2.320	18.61	1	1	4	1
Hornet 4 Drive	21.4	6	258.0	110	3.08	3.215	19.44	1	0	3	1
Hornet Sportabout	18.7	8	360.0	175	3.15	3.440	17.02	0	0	3	2
Valiant	18.1	6	225.0	105	2.76	3.460	20.22	1	0	3	1
Duster 360	14.3	8	360.0	245	3.21	3.570	15.84	0	0	3	4
Merc 240D	24.4	4	146.7	62	3.69	3.190	20.00	1	0	4	2
Merc 230	22.8	4	140.8	95	3.92	3.150	22.90	1	0	4	2
Merc 280	19.2	6	167.6	123	3.92	3.440	18.30	1	0	4	4

油耗量 (mpg) 、排氣量 (disp) 、馬力 (hp) 、汽缸 (cyl) 、化油器數量 (carb) ...

Example

Use dataset mtcars in r

$$H_0 : \beta_2 = \beta_3 = 0$$

$$H_a : \text{at least one } \beta_j \neq 0, j = 2,3$$

Full model : $\text{mpg} = \beta_0 + \beta_1 \text{ disp} + \beta_2 \text{ carb} + \beta_3 \text{ hp} + \beta_4 \text{ cyl}$

Reduced model: $\text{mpg} = \beta_0 + \beta_1 \text{ disp} + \beta_4 \text{ cyl}$

Example

```
#fit full model
```

```
model_full <- lm(mpg ~ disp + carb + hp + cyl, data = mtcars)
```

```
#fit reduced model
```

```
model_reduced <- lm(mpg ~ disp + carb, data = mtcars)
```

```
#perform ANOVA to test for differences in models
```

```
anova(model_reduced, model_full)
```

Analysis of Variance Table

Model 1: mpg ~ disp + carb

Model 2: mpg ~ disp + carb + hp + cyl

	Res.Df	RSS	Df	Sum of Sq	F	Pr(>F)
1	29	254.82				
2	27	238.71	2	16.113	0.9113	0.414

06.

Diagnostics



Multiple linear Regression

Let the model be

$$y_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \cdots + \beta_p x_{ip} + \varepsilon_i, \quad i = 1, \dots, n.$$

Three assumptions are

1. $E(\varepsilon_i) = 0, \quad \forall i$
2. $\text{var}(\varepsilon_i) = \sigma^2, \quad \forall i$
3. $\text{cov}(\varepsilon_i, \varepsilon_j) = 0, \quad \forall i \neq j$
- *4. $\varepsilon_i \sim N(0, \sigma^2)$

Diagnostics

- Constant variance: **Bartlette's Test**

H_0 : the variance is constant

H_a : the variance isn't constant

- Normality assumption: **Anderson-Darling Test** and **Kolmogorov-Smirnov Test**

H_0 : the errors follow a normal distribution

H_a : the errors don't follow a normal distribution

- Independent assumption: **Durbin-Watson test**

H_0 : $\rho = 0$

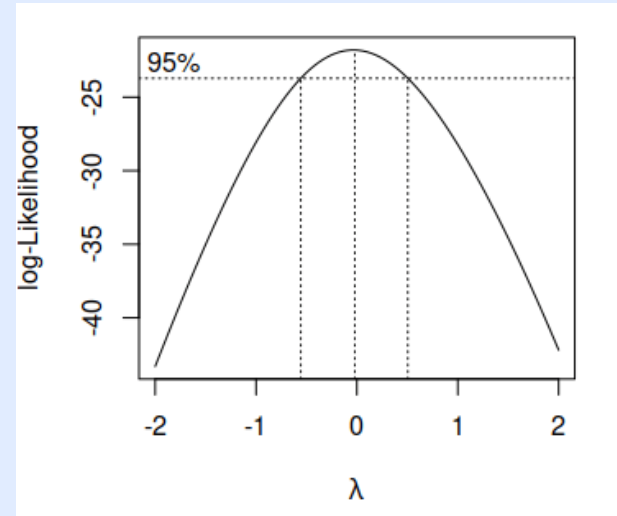
H_a : $\rho > 0$

Diagnostics

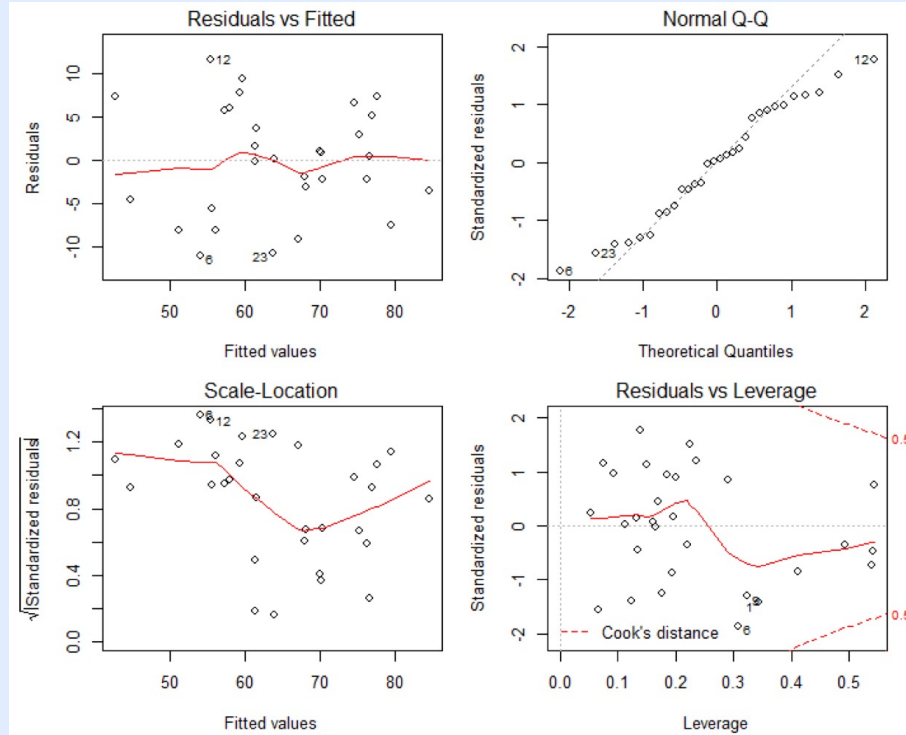
When constant variance assumption is violated

- Transformation on Y
- Box-cox

$$y_i = \begin{cases} \frac{y_i^\lambda - 1}{\lambda}, & \text{if } \lambda \neq 0 \\ \log(y_i), & \text{if } \lambda = 0 \end{cases}$$

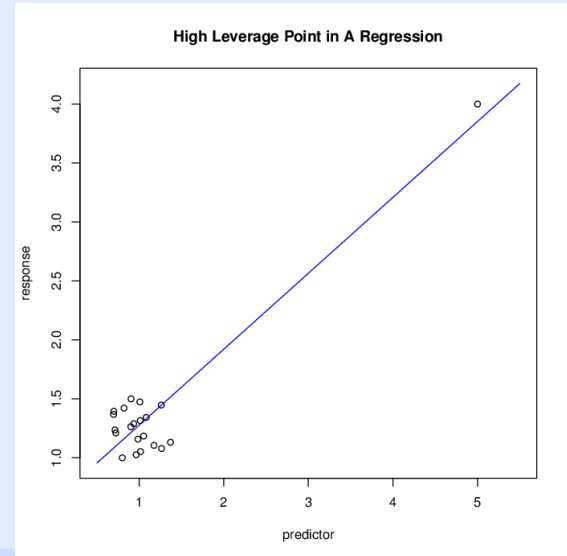


Diagnostics



Diagnostics

- Cook's distance – it is the scaled change in fitted values, which is useful for identifying outliers in the X values. Cook's distance shows the influence of each observation on the fitted response values.

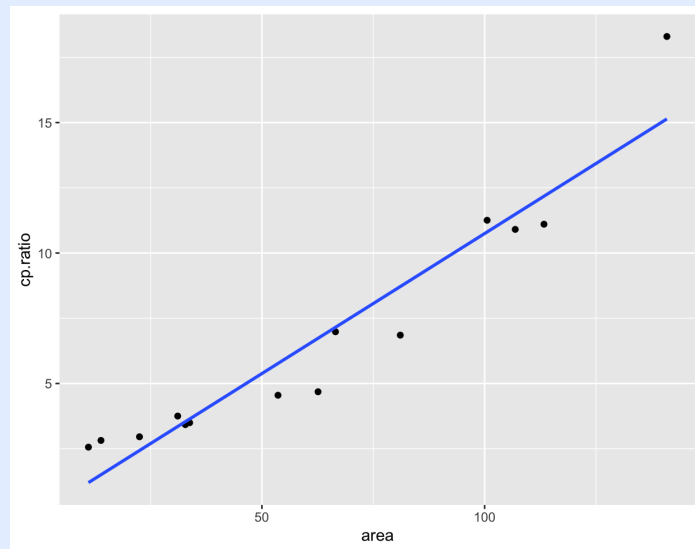
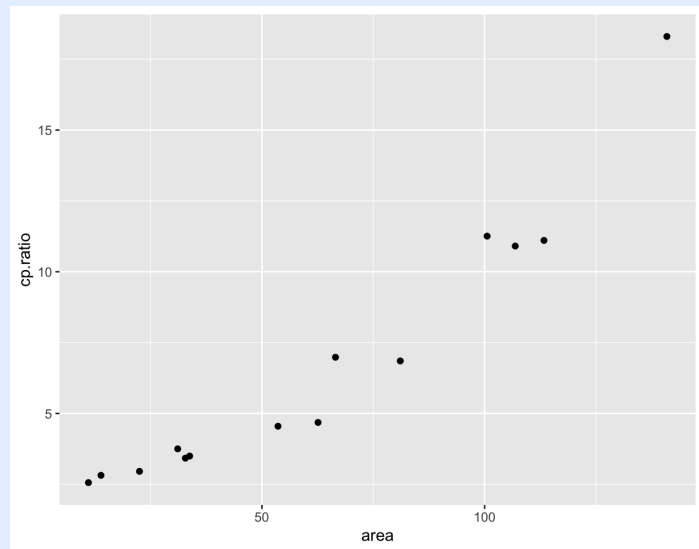


07.

Loess

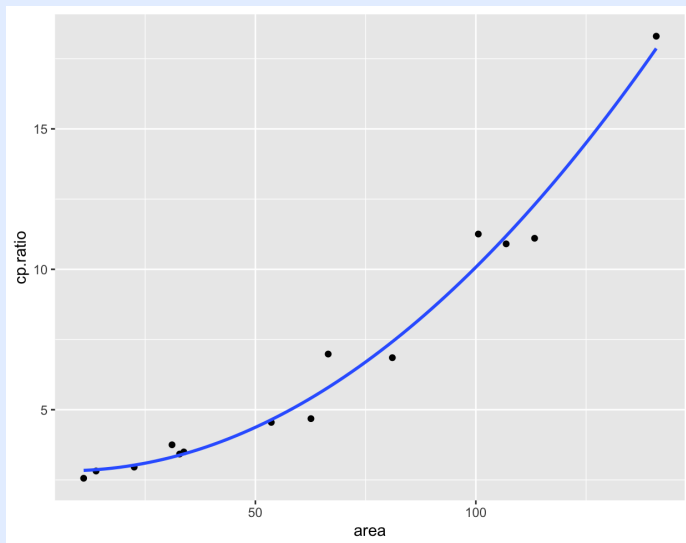


Loess

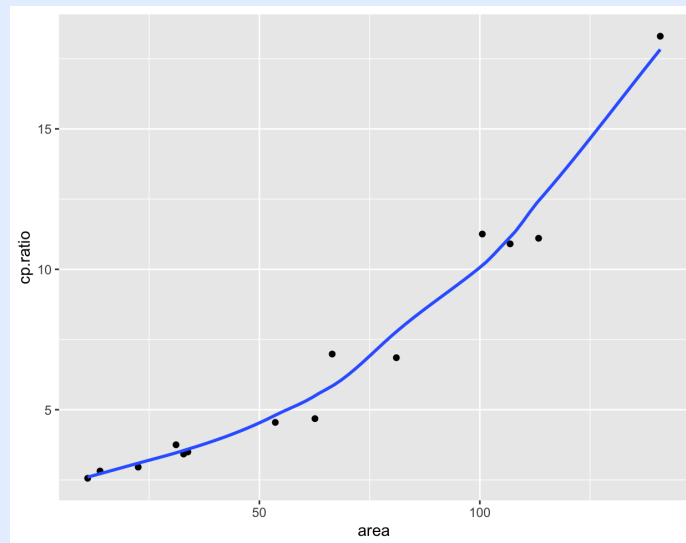


Loess

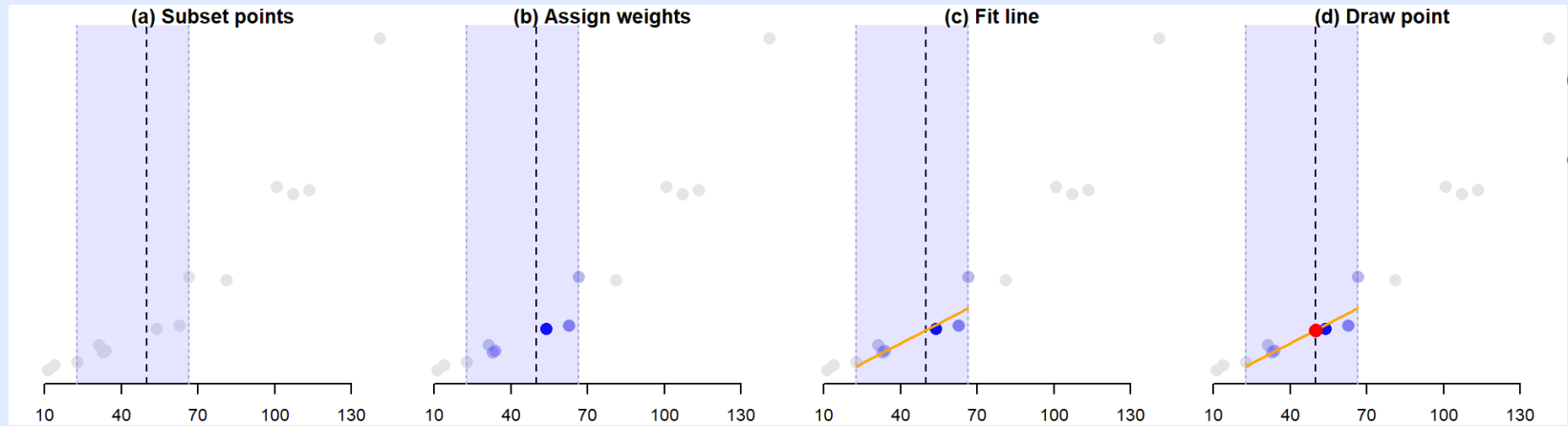
$$y = \beta_0 + \beta_1x + \beta_2x^2$$



Loess



Loess



Loess

- LOESS does not produce a regression function that is represented by a mathematical formula.
- LOESS is somewhat prone to the effects of outliers in the data set

08.

Qualitative Predictors



Qualitative predictors

For a model with qualitative predictors, we need to use the $j-1$ dummy variables (or indicator variables) to show the j categories (levels).

For example, a model with a two-level predictor,

$$Y = \beta_0 + \beta_1 X + \varepsilon$$

where $X = 1$ for level 1, and $X = 0$ for level 2.

- The interpretation of β 's is
- β_0 is the mean for base level (level 2).
- β_1 is the mean difference between the two levels (level 1- level 2).

Qualitative predictors

For a model with a three-level predictor,

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \varepsilon$$

where

$X_1 = 1$ and $X_2 = 0$ for level 1,

$X_1 = 0$ and $X_2 = 1$ for level 2,

$X_1 = 0$ and $X_2 = 0$ for level 3.

The interpretation of β 's is

- β_0 is the mean for **base level** (level 3).
- β_1 is the mean difference between the first and the third levels (level 1- level 3).
- β_2 is the mean difference between the second and the third levels (level 2-level 3).

09.

Interaction Model



An interaction Model with Quantitative Predictors

For a first-order model, the graph of $E(Y)$ against any one predictor, given that the other variables are fixed, is a set of parallel straight lines.

It means that the relationship between $E(Y)$ and any one predictor does not depend on the values of the other variables in the first-order model.

When the relationship between $E(Y)$ and any one predictor depends on the values of the remaining variables, **the interaction model includes the interaction (cross-product) term for expressing the dependence.** For example, an interaction model with two quantitative predictors X_1 , X_2 is given by

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_1 X_2 + \varepsilon$$

An interaction Model with Quantitative Predictors

- Once the importance of interaction has been determined, do not conduct the t test on the model coefficients of the first-order terms.
- Regardless of the magnitude of the associated p-value for testing the model coefficients, the first-order terms should be included in the model.
- Since we rarely know priori whether or not two predictors interact, we will need to fit and test the interaction term to determine its importance.

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_1 X_2 + \varepsilon$$



10.

Model Selection

Introduction

When lots of predictors are considered, the problem of deciding which predictors to include in a multiple regression model is a critical issue.

The variable screening methods **objectively** determine the most important predictors and the least important predictors. Two common variable screening procedures are

- Stepwise Regression
- All-Possible-Regressions Selection

Stepwise Regression

To conduct the stepwise regression, first identify the response variable and k potential predictors.

S1.

Fit all possible simple linear regression models, $Y = \beta_0 + \beta_1 X + \varepsilon$, and conduct the t test for the single parameter β_1 . **Retain the predictor X_1 produces the largest (absolute) t -value.**

S2.

Fit all two-variable models, $Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \varepsilon$, containing X_1 and each of the other $k-1$ predictors, and retain the predictor X_2 having the largest (absolute) t -value. **Go back to check the t -value of $\hat{\beta}_1$ after X_2 adding to the model.** If the t -value is nonsignificant, then remove X_1 and search for another predictor yielding the most significant t -value in the presence of X_2 .

Stepwise Regression

To conduct the stepwise regression, first identify the response variable and k potential predictors.

S3.

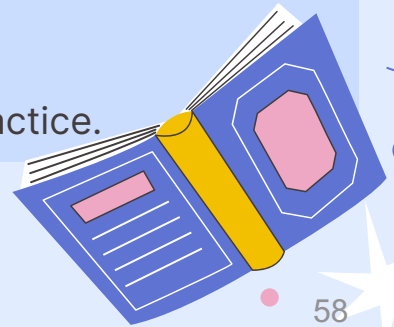
Fit all three-variable models using X_1 , X_2 and each of the $k-2$ remaining predictors, $Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_3 + \varepsilon$. Include the predictor with the largest t -value, and recheck the t -values of X_1 and X_2 after adding X_3 . Replace the predictors having nonsignificant t -values, and continue the procedure until no further predictors yielding significant t -values in the presence of the variables already on the model.

The stepwise regression may **raise the probabilities of Type I error or Type II error due to a large number of t tests**, and it only enters the first-order and main effect terms as candidate variables in the model.

Model selection

Note that

- Besides stepwise regression, **forward selection** and **backward elimination** are two methods for selecting the most important predictors.
- Forward selection is nearly identical to stepwise procedure, except that the forward selection **do not recheck the t-values**.
- Stepwise regression is usually preferred to forward selection in practice.



Model selection

- Backward elimination initially fits the model with k predictors, and drops the predictor having the smallest t -value. The procedure continues until no further nonsignificant predictor is dropped.
- Backward elimination has an advantage when at least one of candidate predictors is a **qualitative variable at or more levels**.
- The disadvantage of backward elimination is to require a sufficiently **large number of data points to fit the initial model**.
- The procedures of stepwise, forward and backward may yield **different results**.

11.

Selection Procedure



Selection Procedure

All-possible-regressions selection procedure considers all possible regression models given the set of potential predictors.

Stepwise regression and all-possible-regressions selection procedures are **different with respect to the criteria** for selecting the important predictors. Four criteria for all-possible-regressions selection procedures are discussed.

- R^2
- Adjusted R^2 or MSE
- C_p
- PRESS (Prediction Residual Error Sum of Squares)



Selection Procedure

01. R^2

The multiple coefficient of determination is defined as

$$R^2 = 1 - \frac{SSE}{SSTO}$$

$$0 \leq R^2 \leq 1$$

The main purpose of the R^2 criterion is to find a subset model with **the larger R^2 nearly as good as the model with all of predictors.**

The measure of R^2 is useful for predicting Y only if the sample contains substantially more data points than the number of model parameters.

Selection Procedure

02. Adjusted R^2 or MSE Criterion

The adjusted R^2 can be expressed by

$$R_a^2 = 1 - \left(\frac{n-1}{n-(k+1)} \right) \left[\frac{SSE}{SSTO} \right] = 1 - \left(\frac{n-1}{n-(k+1)} \right) (1 - R^2)$$

Unlike R^2 , R_a^2 takes into account **the sample size and the number of model parameters**. Note that $R_a^2 \leq R^2$, and the R_a^2 may be negative. Both R^2 and R_a^2 are statistics, **and it is dangerous to assess the usefulness of the model depending only on R^2 and R_a^2** .

We search for the model with higher value of R_a^2 or lower value of MSE.

Selection Procedure

03. C_p Criterion

The C_p is expressed as

$$C_p = \frac{SSE_p}{MSE_k} + 2(p + 1) - n$$

The C_p criterion is to find a subset model with a small value of C_p and C_p near $p+1$.

04. PRESS Criterion

The PRESS(Prediction Residual Error Sum of Squares) is calculated by

$$PRESS = \sum_{i=1}^n [Y_i - \hat{Y}_{(i)}]^2$$

Where $\hat{Y}_{(i)}$ is the predicted value for the i th observation when the model is fit by deleting the i th observation. Models with **small PRESS values are considered good candidate models.**

Selection Procedure

Note that two popular alternative criteria are Akaike's Information Criterion (AIC) and Bayesian Information Criterion (BIC), where

$$AIC = n \ln(SSR) - n \ln(n) + 2p$$

$$BIC = n \ln(SSR) + p \ln(n)$$

- BIC is based on Bayesian statistics and is an information criterion similar to AIC. However, the penalty term $\ln(n)$ in BIC is larger than the 2 in AIC, so it tends to favor simpler models during model selection.
- For $n \geq 8$, BIC is larger than AIC. **We search for the models with small values of AIC, or BIC.**



12.

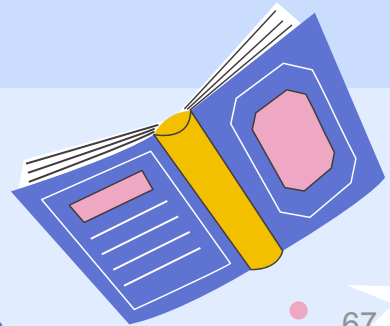
Multicollinearity

Multicollinearity

When the **predictor variables** are highly correlated among themselves, it is called **multicollinearity**.

Its effects are

- Effects on regression coefficients
- Effects on variability of the estimated regression coefficients
- Effects on fitted values and predictions
- Effects on simultaneous tests of regression coefficients



Multicollinearity

The indicators of multicollinearity in regression are as follow:

- There are **significant correlations** between pairs of predictors in the model.
- The **overall F test is significant**, but all(or nearly all) of **individual t-test for the β parameters are nonsignificant**.
- The signs in the estimated parameter are **opposite** from what are expected
- A variance inflation factor (VIF) for a β parameter, where $(VIF)_i = 1/(1 - R_i^2)$, and R_i^2 is the multiple coefficient of determination for the model that regression the predictor X_i on the remaining predictors $X_1, \dots, X_{i-1}, X_{i+1}, \dots, X_k$.

Multicollinearity

Some Solutions for Multicollinearity

- Drop one or more correlated predictors using screening procedures.
- Use the **ridge regression** to estimate the β parameters.
- Code the predictors in **polynomial regression** models.

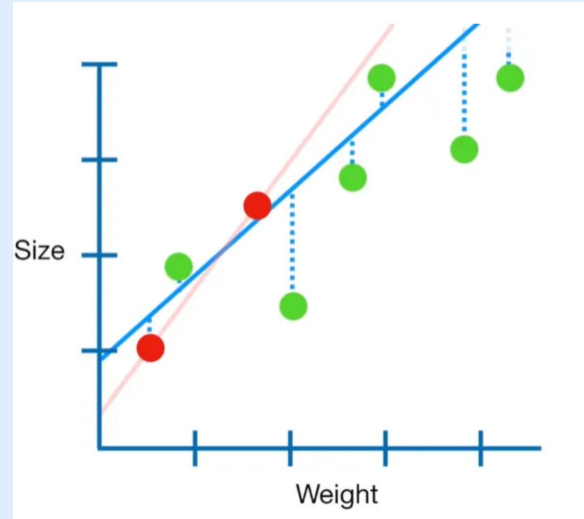
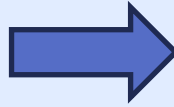
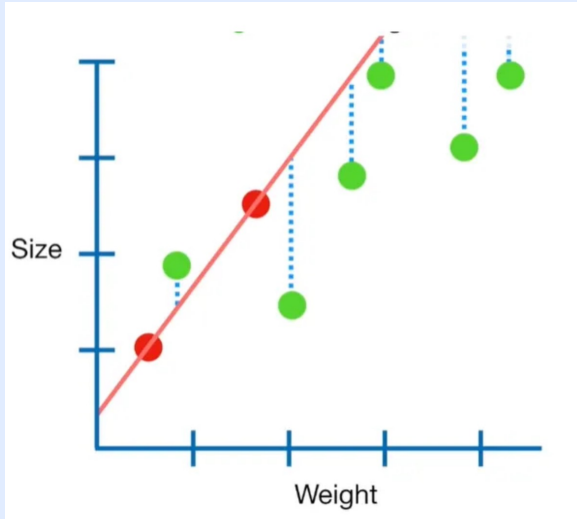
General rule of thumb:

VIF > 4: further investigation.

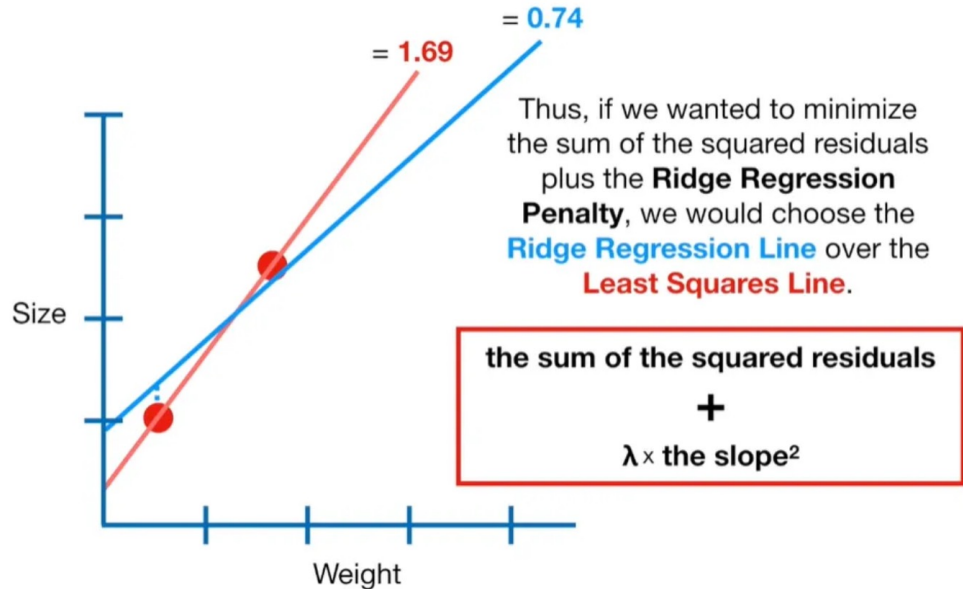
VIF > 10: serious multicollinearity requiring correction.

Ridge Regression

Is a regularization technique used in regression analysis to mitigate multicollinearity and overfitting in models with a large number of predictors



Ridge Regression



The objective function of ridge regression can be written as:

$$\text{minimize } \left(\sum_{i=1}^n (y_i - \beta_0 - \sum_{j=1}^p \beta_j x_{ij})^2 + \lambda \sum_{j=1}^p \beta_j^2 \right)$$

Note that the estimators of the β parameters are biased in ridge regression, but they have smaller standard errors than unbiased estimators by least squares method.

Example – R code

```
data(mtcars)
head(mtcars, 10)
```

	mpg	cyl	disp	hp	drat	wt	qsec	vs	am	gear	carb
Mazda RX4	21.0	6	160.0	110	3.90	2.620	16.46	0	1	4	4
Mazda RX4 Wag	21.0	6	160.0	110	3.90	2.875	17.02	0	1	4	4
Datsun 710	22.8	4	108.0	93	3.85	2.320	18.61	1	1	4	1
Hornet 4 Drive	21.4	6	258.0	110	3.08	3.215	19.44	1	0	3	1
Hornet Sportabout	18.7	8	360.0	175	3.15	3.440	17.02	0	0	3	2
Valiant	18.1	6	225.0	105	2.76	3.460	20.22	1	0	3	1
Duster 360	14.3	8	360.0	245	3.21	3.570	15.84	0	0	3	4
Merc 240D	24.4	4	146.7	62	3.69	3.190	20.00	1	0	4	2
Merc 230	22.8	4	140.8	95	3.92	3.150	22.90	1	0	4	2
Merc 280	19.2	6	167.6	123	3.92	3.440	18.30	1	0	4	4

油耗量 (mpg) 、氣缸數 (cyl) 、排氣量 (disp) 、馬力 (hp) 、車體重量 (wt) 、加速時間 (qsec) ...

Example – R code

以lm()檢驗是否可以用車體重量 (wt) 預測油耗量 (mpg)

Call:

```
lm(formula = mpg ~ wt, data = mtcars)
```

Residuals:

Min	1Q	Median	3Q	Max
-4.5432	-2.3647	-0.1252	1.4096	6.8727

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	37.2851	1.8776	19.858	< 2e-16 ***
wt	-5.3445	0.5591	-9.559	<u>1.29e-10</u> ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 3.046 on 30 degrees of freedom

Multiple R-squared: 0.7528, Adjusted R-squared: 0.7446

F-statistic: 91.38 on 1 and 30 DF, p-value: 1.294e-10

Example – R code

以lm()檢驗是否可以用排氣量 (disp) 預測油耗量 (mpg)

```
Call:
lm(formula = mpg ~ disp, data = mtcars)

Residuals:
    Min       1Q   Median       3Q      Max
-4.8922 -2.2022 -0.9631  1.6272  7.2305

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)  29.599855   1.229720  24.070  < 2e-16 ***
disp        -0.041215   0.004712  -8.747  9.38e-10 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 3.251 on 30 degrees of freedom
Multiple R-squared:  0.7183,    Adjusted R-squared:  0.709
F-statistic: 76.51 on 1 and 30 DF,  p-value: 9.38e-10
```

Example – R code

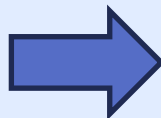
以車體重量 (wt) 和排氣量 (disp) 來預測油耗量 (mpg)

```
Call:
lm(formula = mpg ~ disp + wt, data = mtcars)

Residuals:
    Min       1Q   Median       3Q      Max
-3.4087 -2.3243 -0.7683  1.7721  6.3484

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) 34.96055    2.16454  16.151 4.91e-16 ***
disp       -0.01773    0.00919  -1.929  0.06362 .
wt         -3.35082    1.16413  -2.878  0.00743 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 2.917 on 29 degrees of freedom
Multiple R-squared:  0.7809,    Adjusted R-squared:  0.7658
F-statistic: 51.69 on 2 and 29 DF,  p-value: 2.744e-10
```



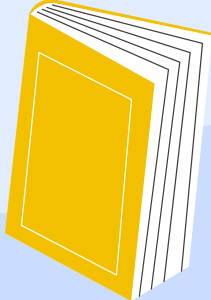
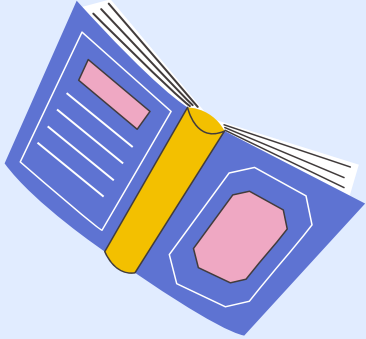
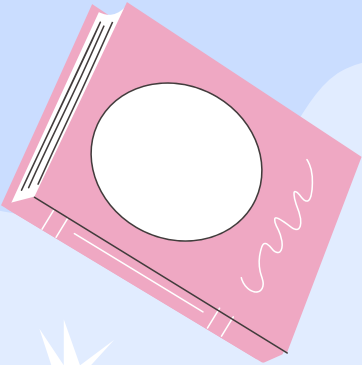
```
model_m_2 %>% vif()
```

disp	wt
4.728319	4.728319

VIF > 4: further investigation.

高排氣量可能意味著更大的車體重量：通常情況下，具有較高排氣量的引擎需要更多的材料和部件來支撐其運作，這可能會增加整車的重量。較大的引擎通常也需要更大的車身來容納，這也可能會導致整體車輛重量的增加。

By ChatGTP



Thanks!

CRÉDITOS: Esta plantilla para presentaciones es una creación de **Slidesgo**, e incluye iconos de **Flaticon**, infografías e imágenes de **Freepik** y contenido de **Sandra Medina**